Parallel Computing

Interconnection Networks for Parallel Computers

Why interconnection network?

- Affects the scalability and cost of the system
  - How large of a system can you build?
  - How easily can you add more processors?
- Affects performance and energy efficiency
  - How fast can processors, caches, and memory communicate?
  - How long are the latencies to memory?
  - How much energy is spent on communication?
- Affects reliability and security
  - Can you guarantee messages are delivered or your protocol works?

Interconnection network basic

- Topology
  - Specifies the way switches are wired
  - Affects routing, reliability, throughput, latency, building ease
- Routing (algorithm)
  - How does a message get from source to destination
  - Static or adaptive
- Buffering and Flow Control
  - What do we store within the routers & links?
    - Entire packets, parts of packets, etc?
  - How do we throttle during oversubscription?
  - Tightly coupled with routing strategy

Terminology

- Network interface
  - Module that connects endpoints (e.g. processors) to network
  - Decouples computation/communication
- Link
  - Bundle of wires that carry a signal
- Switch/router
  - Connects fixed number of input channels to fixed number of output channels
- Channel
  - A single logical connection between routers/switches

Terminology

- Node
  - A router/switch within a network
- Message
  - Unit of transfer for network's clients (processors, memory)
- Packet
  - Unit of transfer for network
- Flit
  - Flow control digit
  - Unit of flow control within network

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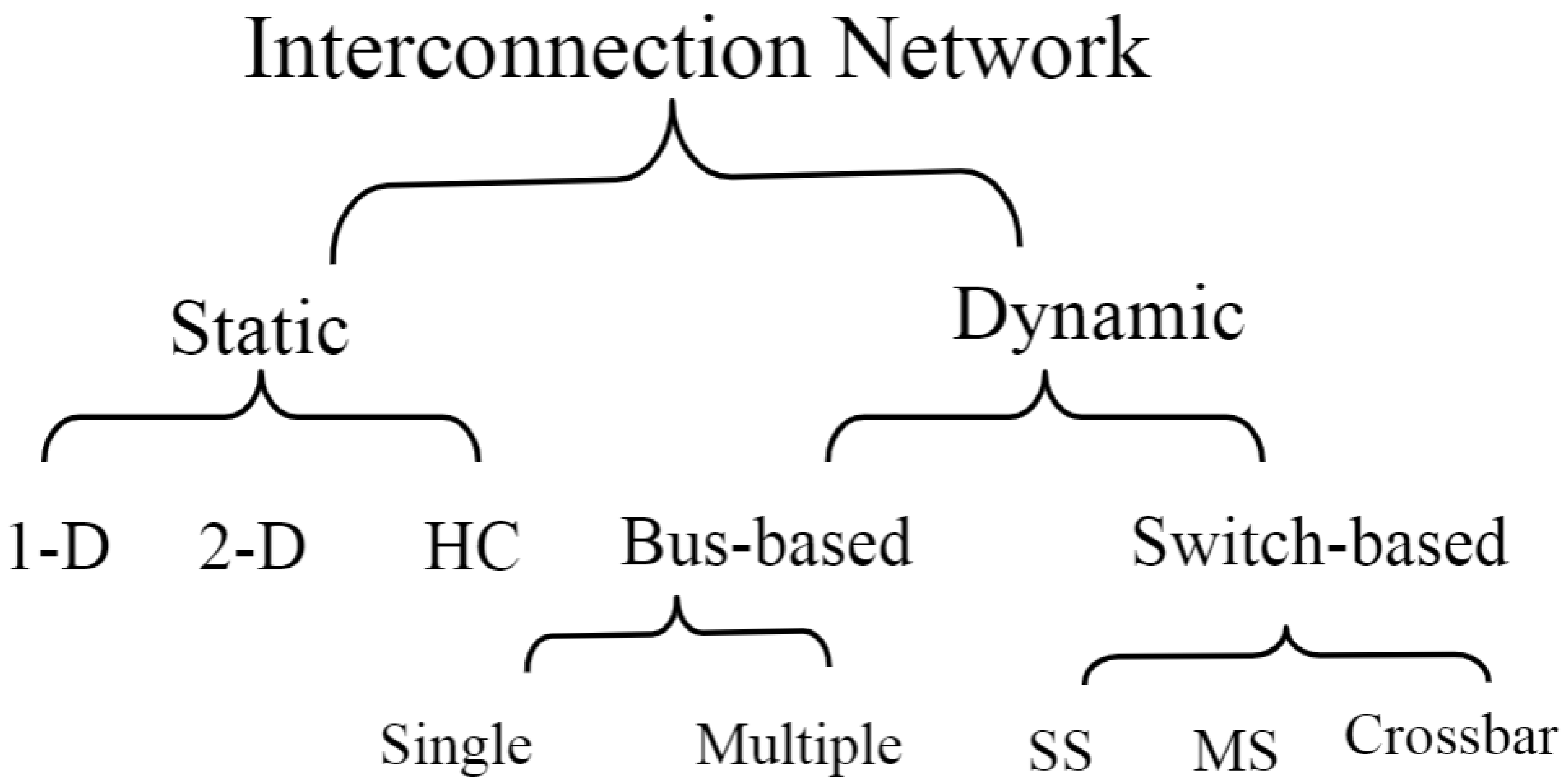
Interconnection Networks for Parallel Computers



hypercube (HC)

Single Stage (SS)

Multi Stage (MS)

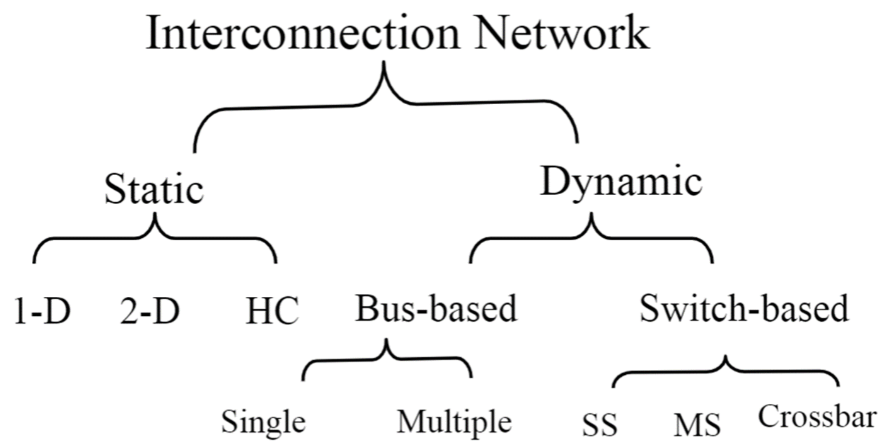
1 Interconnection networks carry data between processors and to memory.

2 Interconnects are made of switches and links (wires, fiber).

3 Interconnects are classified as static or dynamic

4 Static networks consist of point-to-point communication links among processing nodes and are also referred to as direct networks

5 Dynamic networks are built using switches and communication links. Dynamic networks are also referred to as indirect networks



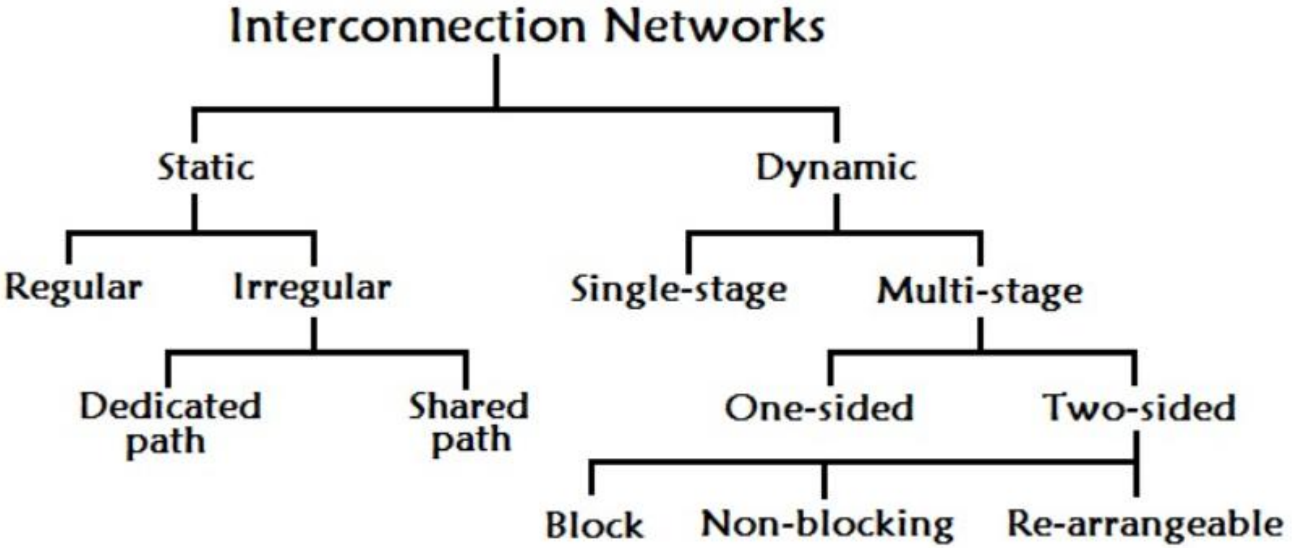
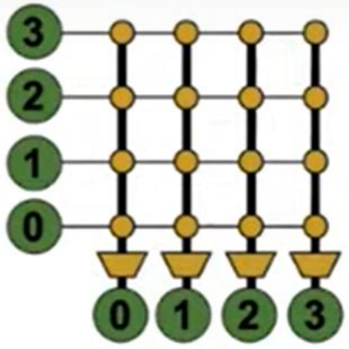
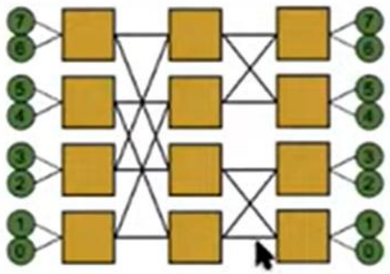
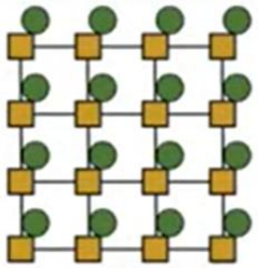


Figure: Classification of Interconnection network





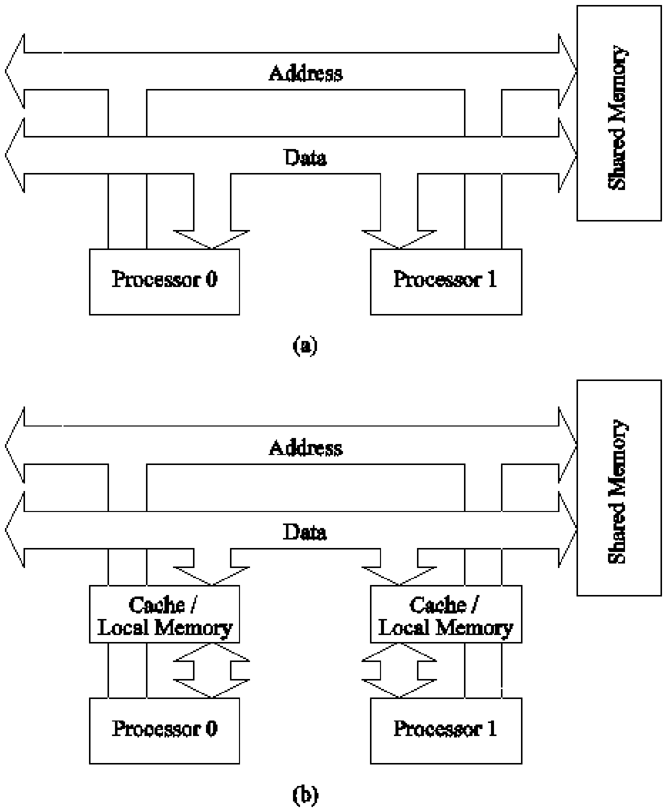


Topology	Crossbar	Multistage Logarith.	Mesh
Direct/Indirect	**Indirect**	**Indirect**	**Direct**
Blocking/ Non-blocking	**Non-blocking**	**Blocking**	**Blocking**
Cost	**$O(N^2)$**	**O(NlogN)**	**O(N)**
Latency	**O(1)**	**O(logN)**	**O(sqrt(N))**

Topology

- Bus (simplest)
- Point-to-point connections (ideal and most costly)
- Crossbar (less costly)
- Ring
- Tree
- Omega
- Hypercube
- Mesh
- Torus
- Butterfly
- ...

Network Topologies: Buses



Bus-based interconnects (a) with no local caches; (b) with local memory/caches.

1 simplest and earliest parallel machines used buses.

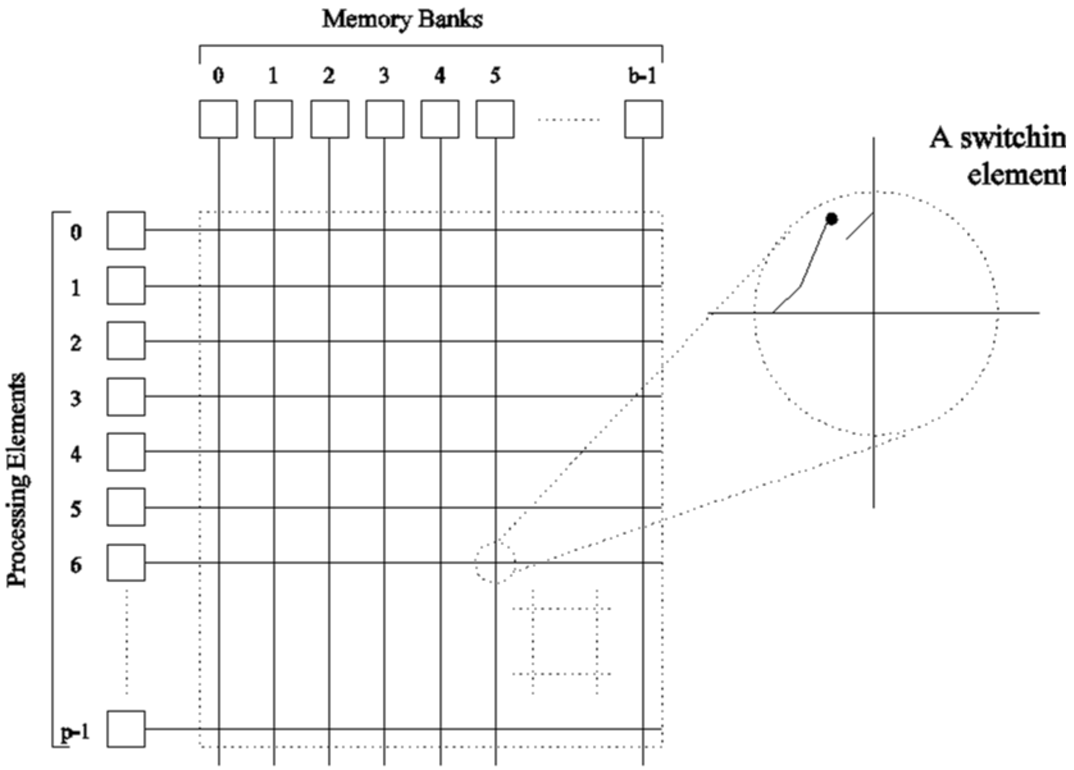
2 All processors access a common bus for exchanging data.

3 The distance between any two nodes is O(1) in a bus. The bus also provides a convenient broadcast media.

4 bandwidth of the shared bus is a major bottleneck

5 Typical bus based machines are limited to dozens of nodes. Sun Enterprise servers and Intel Pentium based shared-bus multiprocessors are examples of such architectures.

Network Topologies: Crossbars



Examples of machines that employ crossbars include the Sun Ultra HPC 10000 and the Fujitsu VPP500.

A crossbar network uses an p×m grid of switches to connect p inputs to m outputs in a non-blocking manner

A completely non-blocking crossbar network connecting p processors to b memory banks.

The cost of a crossbar of p processors grows as O(p^2).

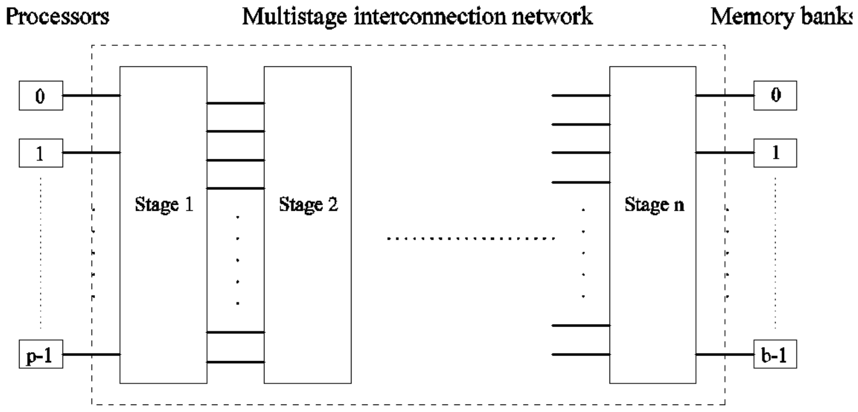
This is generally difficult to scale for large values of p.

Blocking & non-blocking

- A blocking switch prevents connecting more than one input.
- A non-blocking switch allows other concurrent connections from inputs to other outputs.

Network Topologies: Multistage Networks

- Crossbars have excellent performance scalability but poor cost scalability.
- Buses have excellent cost scalability, but poor performance scalability.
- Multistage interconnects strike a compromise between these extremes.



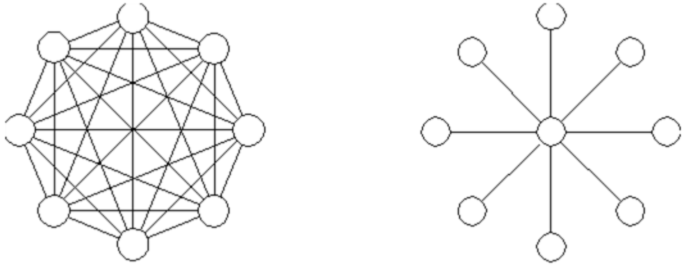
The schematic of a typical multistage interconnection network.

Multistage Omega Network

- The most commonly used multistage interconnects
- This network consists of *log p* stages, where *p* is the number of inputs/outputs
- At each stage, input *i* is connected to output *j* if:

$$j = \begin{cases} 2i, & 0 \leq i \leq p/2 - 1 \\ 2i + 1 - p, & p/2 \leq i \leq p - 1 \end{cases}$$

Network Topologies: Completely Connected and Star Connected Networks



(a) (b)

(a) A completely-connected network of eight nodes;
(b) a star connected network of nine nodes.

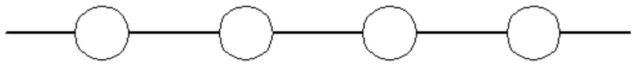
Each processor is connected to every other processor.
The number of links in the network scales as O(p2).
While the performance scales very well, the hardware complexity is not realizable for large values of p.
In this sense, these networks are static counterparts of crossbars.

Every node is connected only to a common node at the center.
Distance between any pair of nodes is O(1). However, the central node becomes a bottleneck.
In this sense, star connected networks are static counterparts of buses.

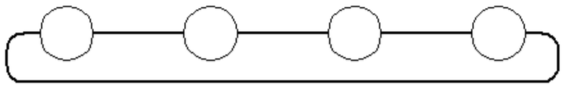
Network Topologies: Linear Arrays, Meshes, and *k-d* Meshes

- In a linear array, each node has two neighbors, one to its left and one to its right. If the nodes at either end are connected, we refer to it as a 1-D torus or a ring.
- A generalization to 2 dimensions has nodes with 4 neighbors, to the north, south, east, and west.
- A further generalization to d dimensions has nodes with 2d neighbors.
- A special case of a d-dimensional mesh is a hypercube. Here, d = log p, where p is the total number of nodes.

Network Topologies: Linear Arrays



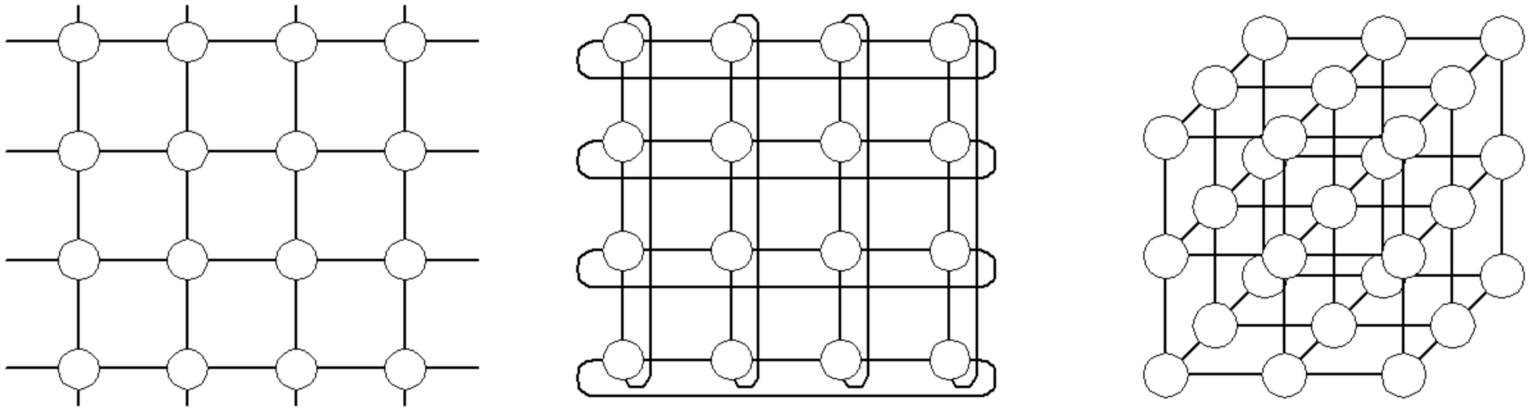
(a)



(b)

Linear arrays: (a) with no wraparound links; (b) with wraparound link.

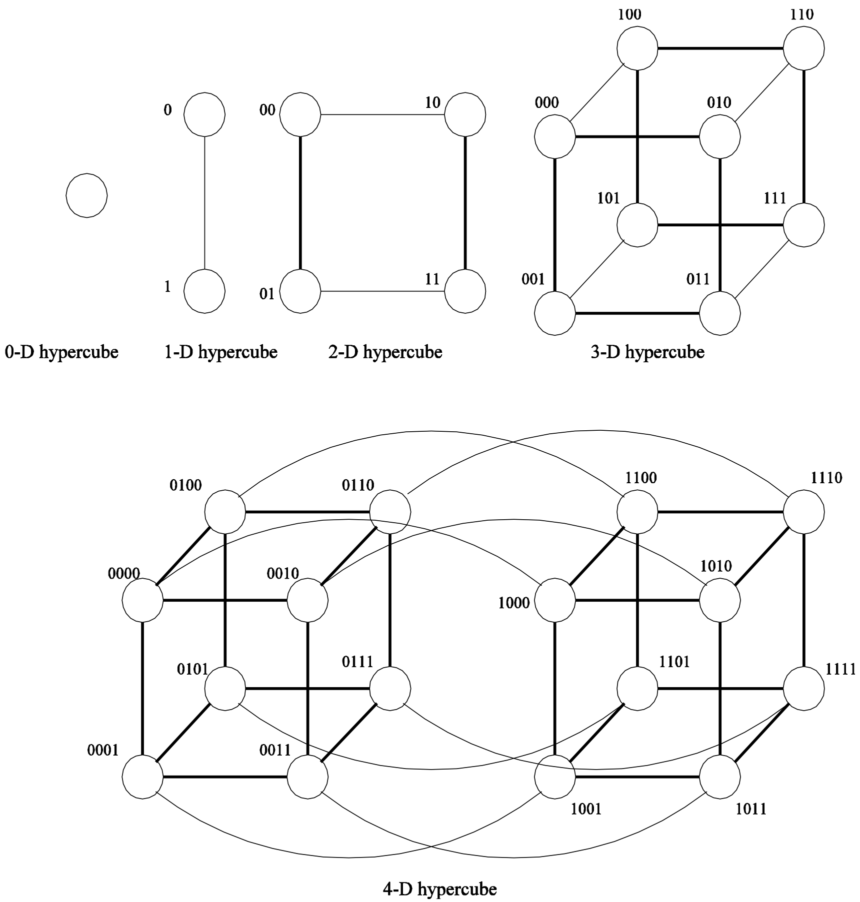
Network Topologies:
Two- and Three Dimensional Meshes



(a) (b) (c)

Two and three dimensional meshes: (a) 2-D mesh with no wraparound; (b) 2-D mesh with wraparound link (2-D torus); and (c) a 3-D mesh with no wraparound.

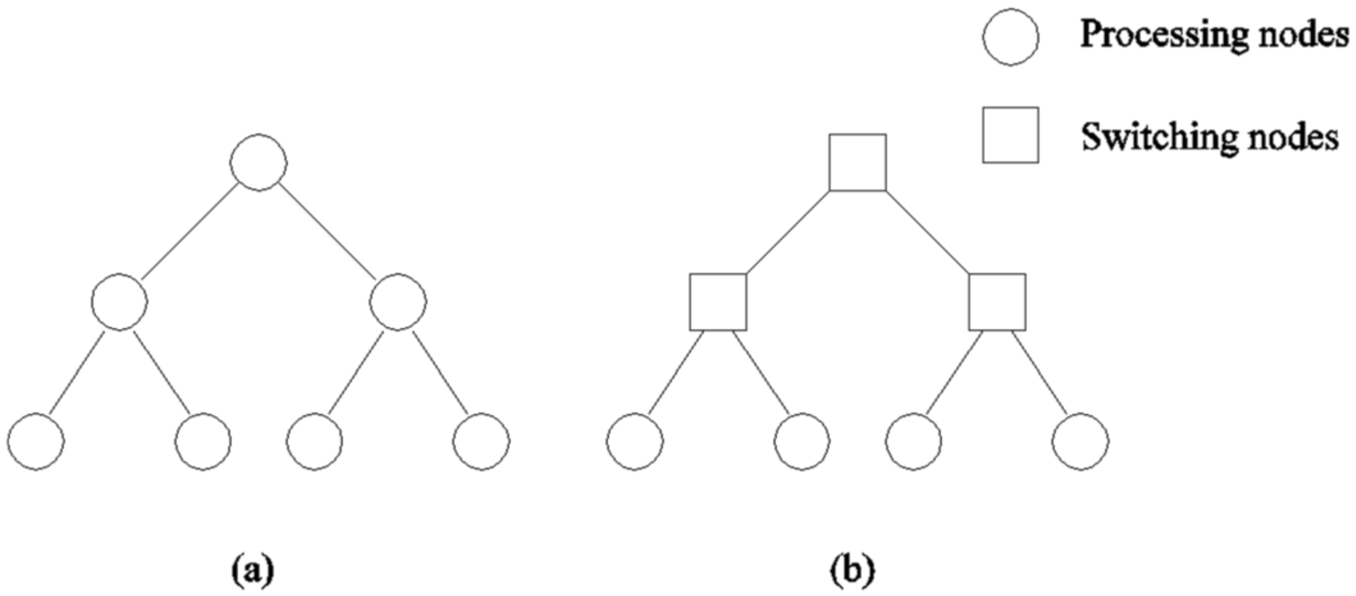
Network Topologies: Hypercubes and their Construction



Construction of hypercubes from hypercubes of lower dimension.

The distance between any two nodes is at most log p.
Each node has log p neighbors.
The distance between two nodes is given by the number of bit positions at which the two nodes differ.

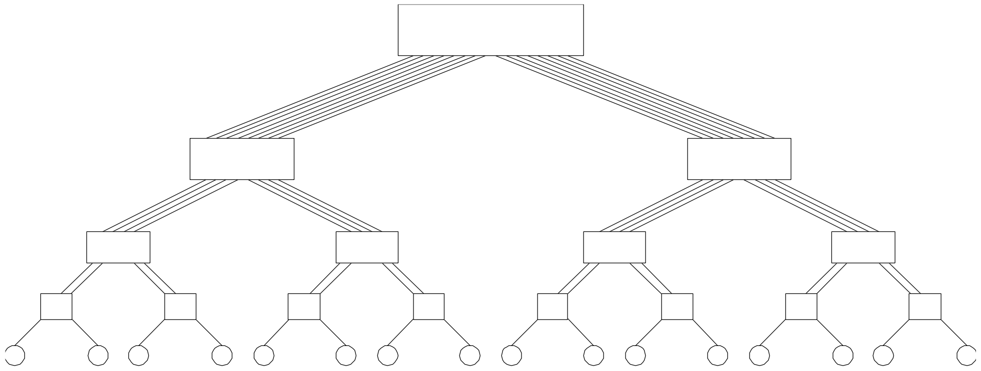
Network Topologies: Tree-Based Networks



Complete binary tree networks: (a) a static tree network; and (b) a dynamic tree network.

The distance between any two nodes is no more than 2logp.
Links higher up the tree potentially carry more traffic than those at the lower levels.
For this reason, a variant called a fat-tree, fattens the links as we go up the tree.
Trees can be laid out in 2D with no wire crossings. This is an attractive property of trees.

Network Topologies: Tree Properties
Fat Trees



A fat tree network of 16 processing nodes.

Evaluating Static Interconnection Networks

- *Diameter:* The distance between the farthest two nodes in the network. The diameter of a linear array is *p − 1*, that of a mesh is *2(− 1)*, that of a tree and hypercube is *log p*, and that of a completely connected network is *O(1)*.
- *Bisection Width:* The minimum number of wires you must cut to divide the network into two equal parts. The bisection width of a linear array and tree is *1*, that of a mesh is , that of a hypercube is *p/2* and that of a completely connected network is $p^2/4$.
- Arc connectivity:…
- *Cost:* The number of links or switches (whichever is asymptotically higher) is a meaningful measure of the cost. However, a number of other factors, such as the ability to layout the network, the length of wires, etc., also factor in to the cost.

Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Completely-connected				
Star				
Complete binary tree				
Linear array				
2-D mesh, no wraparound				
2-D wraparound mesh				
Hypercube				
Wraparound *k*-ary *d*-cube				

Evaluating Static Interconnection Networks

Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Completely-connected	1	$p^2/4$	$p-1$	$p(p-1)/2$
Star	2	1	1	$p-1$
Complete binary tree	$2\log((p+1)/2)$	1	1	$p-1$
Linear array	$p-1$	1	1	$p-1$
2-D mesh, no wraparound	$2(\sqrt{p}-1)$	$\sqrt{p}$	2	$2(p-\sqrt{p})$
2-D wraparound mesh	$2\lfloor\sqrt{p}/2\rfloor$	$2\sqrt{p}$	4	$2p$
Hypercube	$\log p$	$p/2$	$\log p$	$(p\log p)/2$
Wraparound *k*-ary *d*-cube	$d\lfloor k/2\rfloor$	$2k^{d-1}$	$2d$	dp

Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Crossbar				
Omega Network				
Dynamic Tree				

Evaluating Dynamic Interconnection Networks

Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Crossbar	1	p	1	p^2
Omega Network	$\log p$	$p/2$	2	$p/2$
Dynamic Tree	$2 \log p$	1	2	$p - 1$

Evaluating Dynamic Interconnection Networks

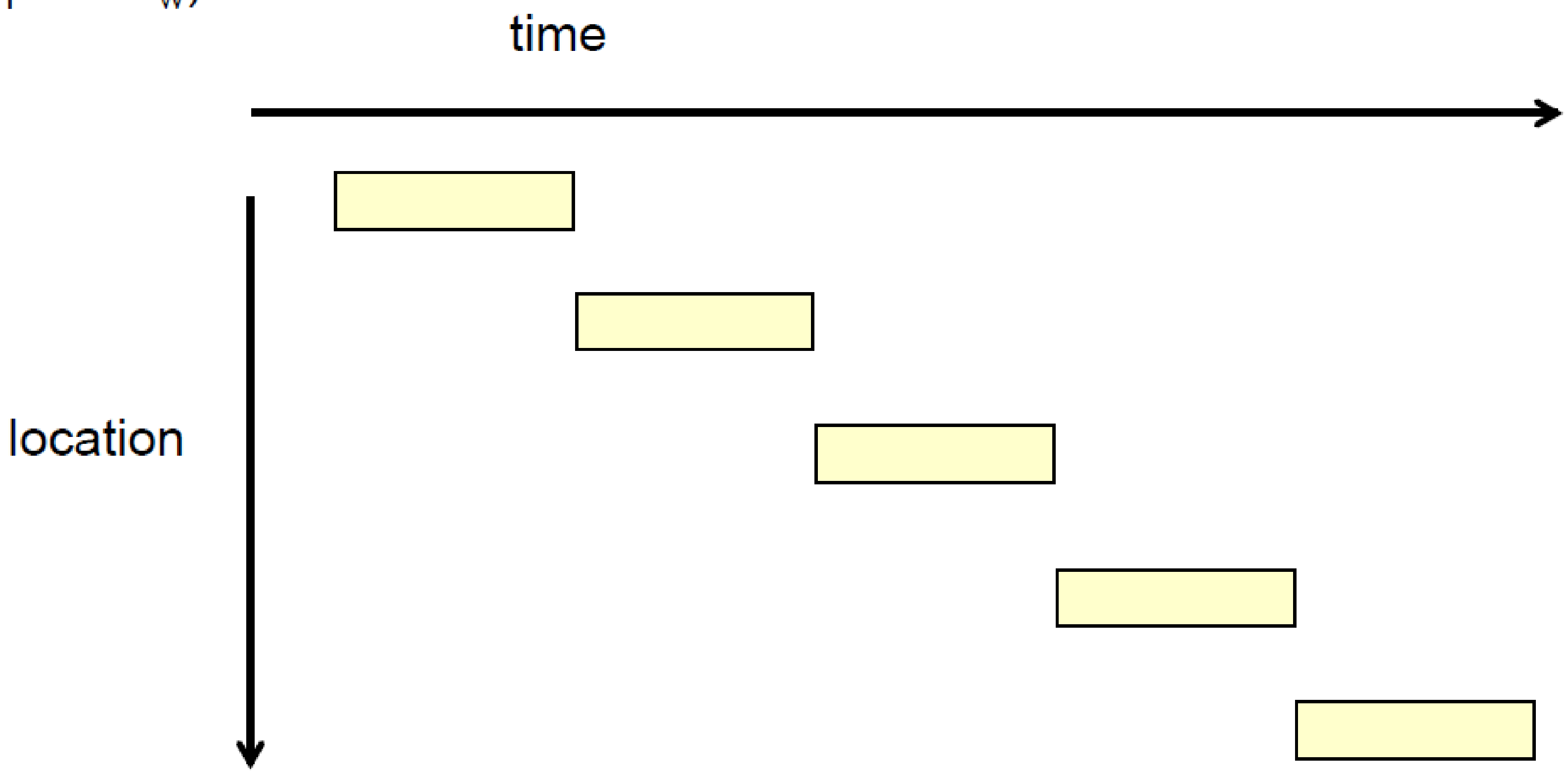
Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Crossbar	1	p	1	p^2
Omega Network	$\log p$	$p/2$	2	$p/2$
Dynamic Tree	$2 \log p$	1	2	$p - 1$

Routing
Store and forward & Cut-through

Store-and-forward routing

- flow-control mechanism at message or packet level
- packet s are transferred one link at a time
- large buffers, high latency
- cost

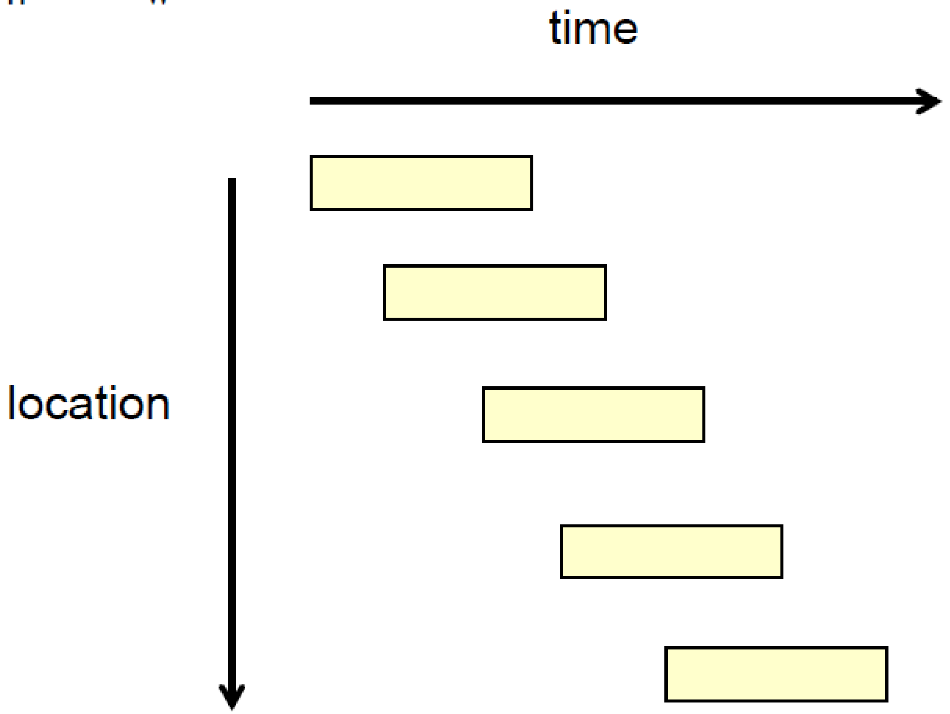
$$t_{SF} = t_s + (t_h + m\ t_w)\ h$$



Cut-through routing

- flow control is per-link and payload transmission is pipelined
- message spread out across multiple links in the network
- small buffers, low latency
- cost

$$t_{CT} = t_s + ht_h + mt_w$$



Further reference

- https://www.youtube.com/watch?v=P_57FZSnrO4